



Realization of energy efficient GF Xtime multiplier using quantum dot cellular automata (QCA) for AES-MixColumn

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Abstract

Recent advances in VLSI technology have led to the introduction of Quantum dot Cellular Automata (QCA) technology as a possible alternative to CMOS technology. This is owing mostly to its tiny feature size, high operating frequency, and low power consumption. During the preliminary research stage, QCA has been used to execute diverse models of combinatorial and sequential circuits, which serve as the fundamental functional components in a wide range of applications. Currently, research is focusing on the implementation of application-oriented architectures using QCA. The motivation behind this research work is to incorporate Galois Field (GF) functions into the AES Mix- Columns operation. We have proposed an Xtime multiplier implemented using QCA technology and analyzed the multiplier using various XOR models of QCA.

Keywords AES MixColumn · Reversible logic · Quantum cost · Cryptography · GF multiplier · Quantum dot cellular automata · QCA designer

1 Introduction

Recent advances in VLSI technology have resulted in the introduction of quantum dot cellular automata (QCA) technology as an alternative to CMOS technology. Quantum-dot

cellular automata (QCA) can build teraflop-frequency integrated circuits (ICs) that have high density and low power consumption. This is achieved by combining QCA cells to form logic gates and wire. QCA-based circuits provide high performance, high reliability, and low power consumption [1–7]. During the preliminary research stage, QCA has been used to execute diverse models of combinatorial and sequential circuits, which serve as the foundational components in a range of applications [8–13]. Currently, scholars are directing their attention toward the execution of application-oriented designs utilizing QCA. Cryptography is the primary enabler of secure data transfer in the information and communication fields. These cryptography methods are classified into different categories based on their operation, key used for encryption and decryption process. A stream cipher and block ciphers are two major categories based on their data and key taken. In block cipher, based on the key methods, it is called symmetric and asymmetric methods, also known as private key and public key methods.

Today's modern secure cryptography algorithms use more complex functions such as permutation, substitution, and mixcolumns. These functions make encryption and decryption more secure compared to simple encryption algorithms.

In the history of cryptography, many algorithms have been proposed and implemented in various applications.

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However, on the other side, due to the improvement in cryptanalysis, most of the algorithms have been broken out. A highly secure block cipher encryption scheme of the Data Encryption Standard has been broken in the DES Challenge (DESCHAL) [14]. Hence, a secure standard algorithm is needed in the security field. Many research works are carried out, and the best was adopted by NIST, USA. The Rijndael Advanced Encryption Standard (AES), which took DES's position as the most excellent secure algorithm, is adopted in 2000. [15, 16]. AES has the following functional modules for secure implementation: AddRoundKey, ShiftRow, Substitution Box, and MixColumn Transformation.

Quantum technologies for realistic computing and simulation have become more apparent and common in recent years. Quantum Dot Cellular Automata (QCA), an alternative to conventional CMOS-based technology, provides a fundamentally new computational basis for designing digital logic circuits using quantum dots contained in the potential well in order to effectively process and transmit information at the nano level. Various methods are proposed to implement the AES encryption and decryption, and different techniques are proposed in the SBox and MixColumn operations. A mixcolumn and SBox are the most critical and power consumption functional modules in AES. Here, we take the Mixcolumn Xtime multiplier module for QCA implementation. The motivation behind this work is to incorporate Galois Field (GF) functions into the AES Mix Columns operation. Objectives of this work are as follows

- Investigate QCA as an alternative to CMOS technology& implement and analyze combinational logic design using QCA.
- Develop and evaluate an Xtime multiplier using various XOR models of QCA, focusing on application-oriented architectures by integrating Galois Field (GF) functions into the AES Mix-Columns operation.

The number of bits in substitution might vary depending on the input format. Most cipher models include substitution, which may be basic or complicated transform-based. A lookup table or logic based on Boolean functions may be used for the implementation. In this paper, a simplified energy and area-efficient basic Xtime multiplier module of the AES Mixcolumn and an inverse mixcolumn have been proposed. This proposed work is carried out in different XOR structures, and its QCA model has been verified. A QCA circuit's size is determined by the total number of cells used in it. Other factors that are affected by circuit complexity include area usage, cell area, large area, and quantum cost. As a result, the performance of these criteria is examined in this proposed work.

The motivation for this work is to address the high computational intensity and power consumption associated with

the MixColumn operation in the AES algorithm, which is a key element of present cryptographic systems. AES is frequently used to secure data, but its complex mathematical operations, particularly the MixColumn operation, require significant computing, energy and resources, making it a bottleneck in cryptographic performance. Traditional CMOS technology is unable to handle these demands efficiently due to physical constraints. The ability of Quantum Dot Cellular Automata (QCA) technology to generate high-density, low-power, and high-frequency integrated circuits makes it an appealing alternative. Using QCA for the Xtime multiplier in the MixColumn operation, this research intends to dramatically boost the efficiency, performance, and reliability of AES implementations, making it a valuable and innovative solutions for cryptography applications.

The remaining sections of the paper are organized in the following way: Section 2 of this paper examines the existing literature on the implementation of advanced encryption standard (AES) and Quantum-dot cellular automata (QCA). Section 3 provides a concise overview of the distinct AES function modules and their respective functionalities. Section 4 provides a comprehensive discussion of the fundamental principles and concepts of QCA. The utilization of various XOR operations in the proposed GF Xtime multiplier is shown in Sect. 5. The discussion on simulation results is shown in Sect. 6. Section 7 outlines a description of the work's conclusion and future scope.

2 Literature survey

AES was developed using a Coprocessor, the FPGA, an ASIC, and SOC, with the primary goal of attaining maximum power efficiency, reduced area requirements, and increased operational speed. The nano design approach employs various design styles, including QCA. The first part of this paper is focused on broad QCA design implementations classified into three general categories: fundamental logic gate design, combinatorial and sequential circuit design, and application-oriented design. The second part discussed the implementation of Xtime multiplier for AES mixcolumn and inverse mixcolumn. Many research investigations have been conducted on implementation of AES MixColumns and QCA separately. However, there is a limited number of works conducted in the field of AES specifically focusing on QCA technology. In [17], the authors have implemented the overall flow of the AES structure in the QCA model. In addition, [18], a detailed implementation of AES Sbox and its energy efficient, area parameters are analyzed in QCA model. In reference [19], a polymorphic multiplication based on $GF(2^4)^2$ and $GF(2^8)$ has been introduced. This multiplication operation is implemented using the pipeline structure model. Despite its high-speed

operation, which accelerates the computational task of addition in the process of multiplication. In [20], authors have conducted a substructure sharing mixcolumn operation, while [21] introduced concept of combined matrix multiplication model. A systolic array multiplication has been discussed in [22] to improve the throughput performance of mixcolumn operations. A vedic multiplier-based parallel multiplication has been proposed by authors in [23] to improve the area of the conventional multiplication.

In [24], authors implemented a low power strategy that involved the utilization of resource sharing. Additionally, the authors in [25] proposed the incorporation of a limited critical route and a less gated architecture. Furthermore, a novel implementation of the mixcolumn operation was proposed in [26], employing a Field-Programmable Gate Array (FPGA) design that incorporated four input Look-Up Tables (LUTs). The dynamic mixcolumn transformation was explored in [27]. The implementation of MixColumn in the aforementioned methods is influenced by architectural considerations, resource sharing principles, and a pipeline model. Several researchers have focused their efforts on modifying the logic gates at the circuit level. This proposed work focuses on the optimization of a logic-based X-time multiplier with respect to gate count.

Many research investigations have been conducted on the utilization of majority voter gates, inverters, and XOR gates in the implementation of combinational logic circuits and sequential circuits. The majority voter gate is commonly employed in Xor gate investigations. However, a few of them focus on the direct implementation of the XOR gate using various methods. In [28], an energy-efficient XOR model for half adder and full adder circuits, aiming to minimize the number of gate counts, while achieving good performance has been presented. Additionally in [29], authors have also proposed an XOR-based adder circuit with similar objectives.

3 Advanced encryption standard

Rijndael AES was accepted as the Advanced Encryption Standard (AES) by the National Institute of Standards and Technology (NIST) in 2001. Rijndael AES has been shown to be more secure than other encryption techniques. The AES adheres to a fixed data block length of 128 bits and provides the option of utilizing a key length of 128, 192, or 256 bits [15]. In contrast, the Rijndael AES algorithm allows for the independent selecting of both the input block length and the key length. The size of this block adhere to a 32-bit multiple, specifically ranging from 128 to 256 bits. The Rijndael cipher exhibits the ability to accommodate diverse permutations of block and key lengths. The rounds are assigned as 10, 12, and 14. The AES algorithm uses key sizes of 128,

192, and 256. Joan and Vincent claimed that the effectiveness of ciphertext can be enhanced by using more lengthy key sizes [16].

The AES comprises two fundamental operations, namely encryption and key generation. The encryption module comprises four distinct functional modules, specifically the SBox, add round key (ARK), ShiftRow (SR), and Mix-column transformation (MCT). The decryption procedure utilizes function modules that are identical and employ the inverse model, namely InvSBox, InvSR, and InvMCT.

The functional blocks of AES that consume a significant amount of power are the SBox and MCT [30].

3.1 AES modules

Sub byte transformation(SBOX): The non-linear sub-byte transformation adds a bijective function to each state matrix model byte. This technique requires preparatory calculation and SBox value retention or real-time combinational logic equation computation. Each byte of the 128-bit input data is structured into a 4-by-4 matrix [15, 16].

Add round key: The AddRoundKey process is an essential step that requires the execution of an XOR operation between the result of the MixColumns transformation and the key expansion part. The key expansion unit produces keys for every round [15, 16].

Shift row: The pattern of leftward shift of rows is suggested. The method of performing a shift row operation on 128-bit data entails the shift of each byte in each row. The main procedure includes the utilization of a permutation operation that serves to introduce an additional level of complexity to the AES round [15, 16].

MixColumn transformation: The MixColumn operation is a non-linear process that employs inter-byte mixing and is acknowledged for its substantial power consumption. The 4x4 matrix is subjected to a constant multiplication with the 4x1 input matrix. The provided information pertains to a polynomial function that is subjected to a modulo operation with respect to $x^4 + 1$. This polynomial function's coefficients are members of the Galois Field (GF) of order 2^2 . The state matrix can serve as a representation of the input for MCT [15, 16, 30].

4 Quantum dot cellular automata cell (QCA cell)

The introduction of QCA was initially proposed by Lent et al. [32]. The QCA nanostructure presents a novel approach for encoding, transmitting, and processing binary information. A collection of small quantum dots with a nano-structured composition are organized to simplify the execution of individual computations [33]. The states of logic in QCA

cells are contingent upon the positioning of electrons, as opposed to the voltage of the cell [32, 34, 35].

4.1 QCA cell

A cell layout is made by mapping the circuit's logic gates to their equivalent QCA cell structures and putting the cell structures on a 2-dimensional surface. Upon placement of the QCA logic gates, arranging other QCA cells on a two-dimensional surface is essential to establish an interconnect between the logic gates and input/output components [36].

QCA cells are classified into two types: type 1 and type 2, which are marked by the location of four points inside the QCA structure. Type 1 and Type 2 refer to two distinct configurations of points within a cell. QCA cells can be oriented in two ways: at an 90° angle or an 45° angle. The standard orientation for cells is 90 degrees. A cell with an 45° orientation can be considered a 90° cell rotated 45° counterclockwise. In the QCA architecture, the communication between individual electrons is facilitated by Colombian repulsion, which replaces the conventional current-switching transistor. A detailed structure of the Type, and Type 2 cells are shown in Fig. 1a, b. A basic majority voter circuit is in Fig. 1c.

4.2 QCA wires

Binary data transmission in a QCA wire is influenced by electrostatic interactions between cells at the input and output. Figure 1d depicts the propagation within a 90° QCA wire. In addition to utilizing a 90° QCA wire, it is also possible to employ a 45° QCA wire, as shown in Fig. 1e. In this instance, the transmission of the binary signal varies between the two polarizations [14].

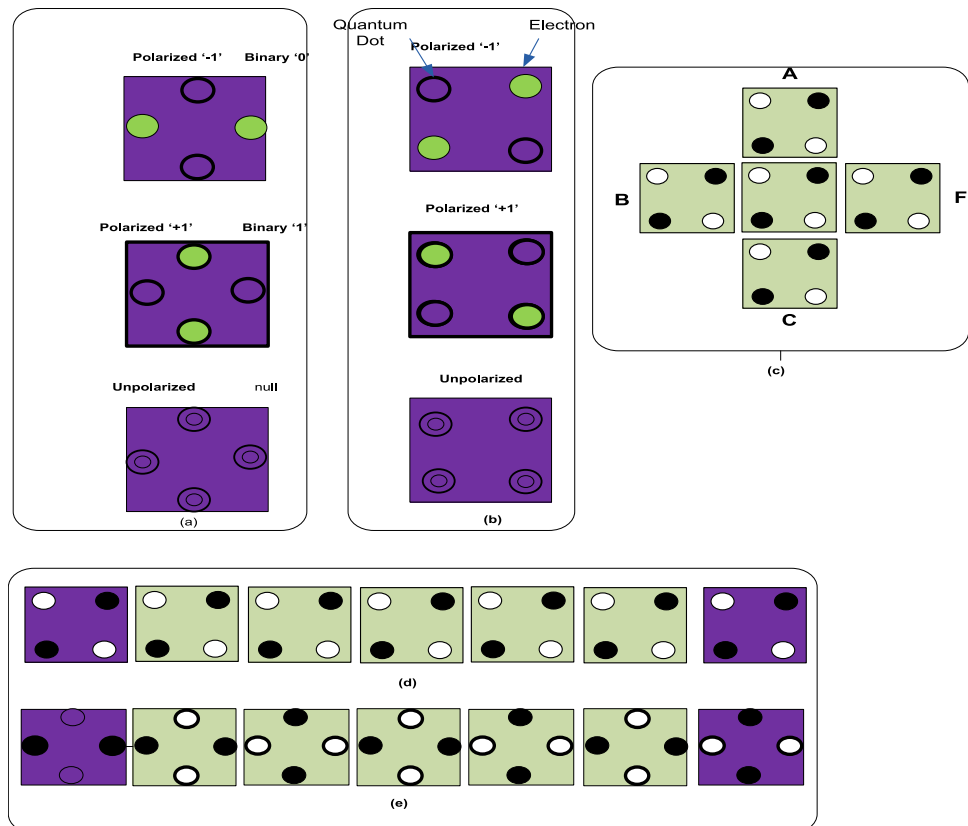
4.3 QCA gates

The QCA technology comprises two fundamental gate types, specifically, the Majority gate and the NOT gate. Indeed, the OR and AND gate are designed from the majority gate. The gate, commonly called the majority gate, generates an output resulting from most of the inputs it receives. The QCA majority gate consists of four terminal cells, with three acting as input terminals and the fourth being used as the output cell [5].

$$F(A, B, C) = AB + BC + CA \quad (1)$$

Eq.(1) It represents a majority gate model, with the inputs shown by parameters A, B, and C.

Fig. 1 a QCA - Type 1, b Type 2 Cells, c Majority Voter, d Type 1 and e Type 2 Wires



The inverter gate is formed by placing its cell in the adjacent corner, resulting in polarization inversion at the opposite level. According to this, an inverter has been designed in two distinct models. The two-cell inverter is a basic model comprising two cells working on the principle of opposing electrons located in opposite corners. Nevertheless, the inverter is subject to signal integrity issues, leading to the selection of a more resilient type of inverter, though at the cost of additional cells [35–37].

The XOR gate is a widely used element in digital circuits such as parity checkers, parity detectors, code converters, Arithmetic logic unit and reversible logic circuits. In addition to the fundamental AND, OR, and Inverter gates, the XOR gate plays a significant role in digital circuit design. The XOR gate is essential to our design, and we have emphasized certain QCA XOR models [38, 39].

4.4 QCA clocking

Similar to clock-based CMOS circuits, the inter-dot barrier of QCA is controlled by the clock signal. The pipeline-based timing mechanism governs the operation of QCA designs that are either combinational or sequential.

A clock serves a dual role of synchronizing and controlling the flow of information, as well as providing the necessary power to operate the circuit, given the absence of an external source for powering cells. The process of the QCA clock can be achieved by regulating potential barriers between quantum dots near one another (Figs. 2, 3).

The pipeline structure is a model consisting of four separate phases, notably the Switch, Hold, Release, and Relax phases. The QCA approach employs four discrete clocks, particularly CLOCK 0, CLOCK 1, CLOCK 2, and CLOCK 3, which are synchronized at 90° intervals.

During the initial stage of the switching process, the QCA cell initiates polarization, and the inter-dot barriers are

incrementally intensified. Ultimately, the QCA cell attains a specific polarization state, which is determined by the polarization states of its adjacent cells. During the second phase of the Hold process, the cell exhibits a high inter-dot barrier, which serves as a fixed polarization to facilitate the activation of the subsequent cell. Subsequently, during the Release and Relax phase, the cells return to an unpolarized state. These two next phases cannot influence the determination of the QCA cell state [5, 38, 39].

4.5 XOR gate-different structure of QCA

The XOR logic gate is a fundamental logic gate utilized in the mix column and inverse mix column operations, which involve the manipulation of two or more inputs [28].

5 Proposed GF Xtime multiplier-logic gates

5.1 MixColumn transformation

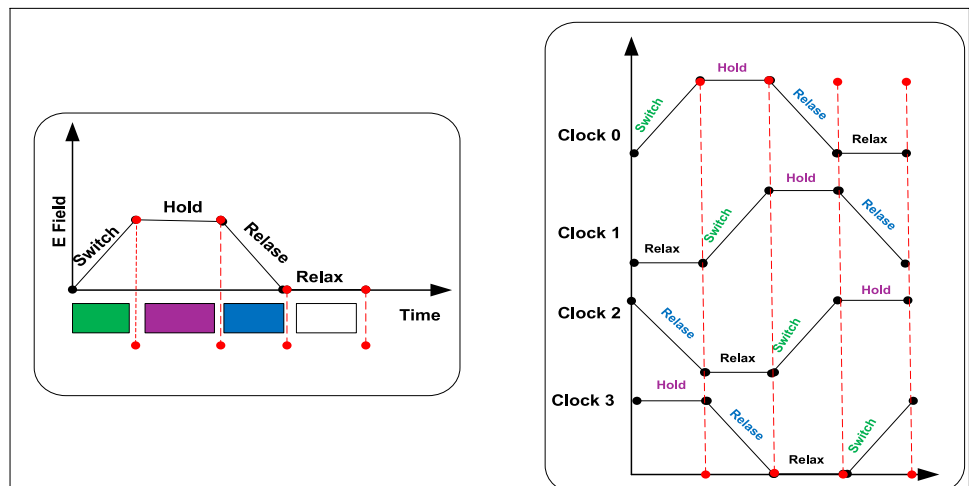
The Mix Column Transformation (MCT) operation is a fundamental component of the AES encryption algorithm, while its inverse, InvMCT, is utilized in the decryption process. The MCT matrix constants for 02 and 03 Xtime multipliers are computed using Eq.(3) which are also described in matrix form Eq.(2).

$$\begin{bmatrix} wc_0 \\ wc_1 \\ wc_2 \\ wc_3 \end{bmatrix} = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} * \begin{bmatrix} xc_0 \\ xc_1 \\ xc_2 \\ xc_3 \end{bmatrix} \quad (2)$$

$$wc_0 = 02 * xc_0 \oplus 03 * xc_1 \oplus 01 * xc_2 \oplus 01 * xc_3 \quad (3a)$$

$$wc_1 = 01 * xc_0 \oplus 02 * xc_1 \oplus 03 * xc_2 \oplus 01 * xc_3 \quad (3b)$$

Fig. 2 QCA - clock model



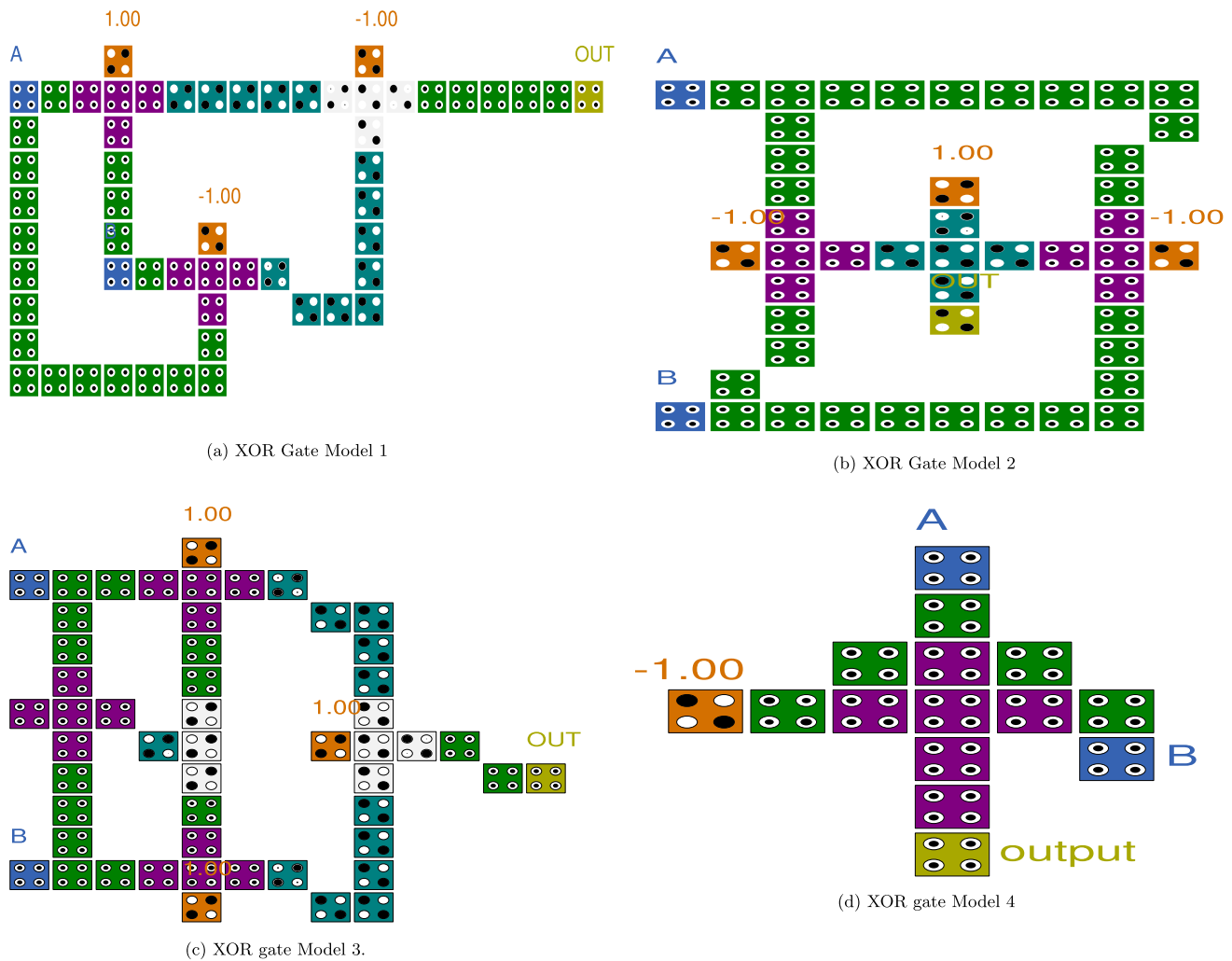


Fig. 3 Different XOR gate model for Xtime multiplier 02

$$wc_2 = 01 * xc_0 \oplus 01 * xc_1 \oplus 02 * xc_2 \oplus 03 * xc_3 \quad (3c)$$

$$wc_3 = 03 * xc_0 \oplus 01 * xc_1 \oplus 01 * xc_2 \oplus 02 * xc_3 \quad (3d)$$

The 02 multiplication method employs a shift operation, while the 03 multiplication method utilizes both the 02 multiplication method and the XOR operation. This work uses simple 02 multipliers and makes 03 multipliers based on 02 multiplication as shown in Eq.(4). The use of 02 multipliers results in a reduction of power and area.

$$\begin{aligned} wc_0 &= 02 * xc_0 \oplus 03 * xc_1 \oplus 01 * xc_2 \oplus 01 * xc_3 \\ &= 02 * xc_0 \oplus (02 * xc_1 \oplus 01 * xc_1) \oplus 01 * xc_2 \oplus 01 * xc_3 \\ &= 02 * (xc_0 \oplus xc_1) \oplus 01 * xc_1 \oplus 01 * xc_2 \oplus 01 * xc_3 \\ &= 02 * (xc_0 \oplus xc_1) \oplus 01 * xc_1 \oplus 01 * xc_2 \oplus 01 * xc_3 \oplus xc_0 \oplus xc_0 \\ &= 02 * (xc_0 \oplus xc_1) \oplus tmp \oplus xc_0 \end{aligned} \quad (4a)$$

similarly, the remaining terms, wc_1, wc_2, wc_3 are written as

$$wc_1 = 02 * (xc_1 \oplus xc_2) \oplus tmp \oplus xc_1 \quad (4b)$$

$$wc_2 = 02 * (xc_2 \oplus xc_3) \oplus tmp \oplus xc_2 \quad (4c)$$

$$wc_3 = 02 * (xc_3 \oplus xc_1) \oplus tmp \oplus xc_3 \quad (4d)$$

According to Eq.(4), the MCT operation can be done by using a 02 Xtime multiplier. The simplified way to use an XOR gate is employed for the implementation of the 02 XTime GF

multiplier. QCA architecture is implemented using a comparable technique by utilizing an isolated XOR architecture to minimize energy consumption. The proposed design was used to investigate the energy consumption of MCT through the implementation of various XOR structures.

The multiplication of 02 is typically performed using the shift addition method, which is known for resulting in increased area and power consumption. The conventional approach for performing 02-multiplication involves utilizing eight flip flops and an eight bit XOR gate to execute eight bit shift operations. The revised binary multiplication algorithm, denoted as 02 Xtime, necessitates a mere three XOR gates shown in Fig. 4.

5.2 Inverse MixColumn

The representation of the matrix multiplication of inverse MixColumn in Rijndael AES is given by an Eq. (5) (Figs. 5, 6)

$$\begin{bmatrix} xc_0 \\ xc_1 \\ xc_2 \\ xc_3 \end{bmatrix} = \begin{bmatrix} 0E & 0B & 0D & 09 \\ 09 & 0E & 0B & 0D \\ 0D & 09 & 0E & 0B \\ 0B & 0D & 09 & 0E \end{bmatrix} * \begin{bmatrix} wc_0 \\ wc_1 \\ wc_2 \\ wc_3 \end{bmatrix} \quad (5)$$

The structure obtained by simplifying Eq.(6) using G.F. polynomial notation is then employed in AES decryption [31].

$$xc_0 = 0E * wc_0 \oplus 0B * wc_1 \oplus 0D * wc_2 \oplus 09 * wc_3 \quad (6a)$$

$$xc_1 = 09 * wc_0 \oplus 0E * wc_1 \oplus 0B * wc_2 \oplus 0D * wc_3 \quad (6b)$$

$$xc_2 = 0D * wc_0 \oplus 09 * wc_1 \oplus 0E * wc_2 \oplus 0B * wc_3 \quad (6c)$$

$$xc_3 = 0B * wc_0 \oplus 0D * wc_1 \oplus 09 * wc_2 \oplus 0E * wc_3 \quad (6d)$$

In Eq.(6), the constant 0E,0B,0D,09 are represented as follows: $0E_{16} = 08 \oplus 04 \oplus 02$; $0B_{16} = 08 \oplus 02 \oplus 01$;

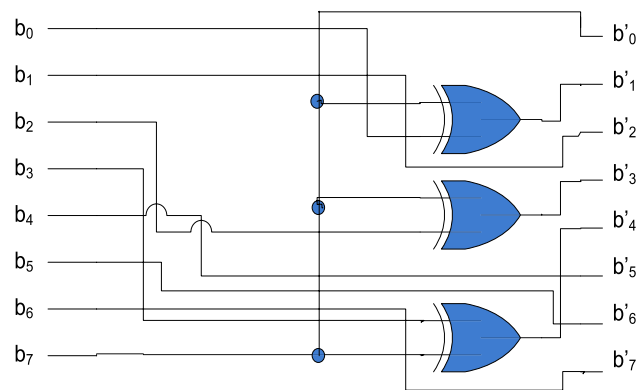


Fig. 4 02 Xtime GF multiplier

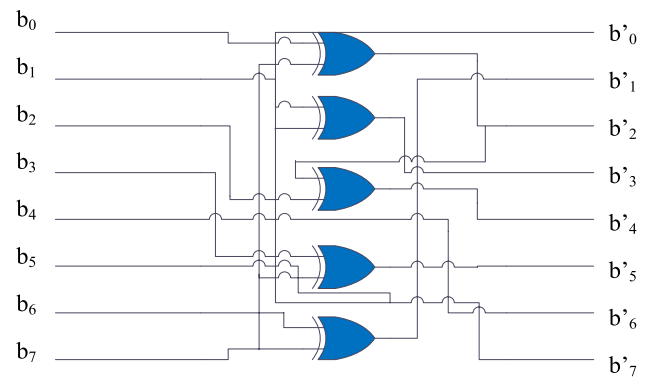


Fig. 5 04 Xtime GF multiplier

$0D_{16} = 08 \oplus 04 \oplus 01$; $09_{16} = 08 \oplus 01$; Finally, the simplified Equation for one 8-bit data is

$$\begin{aligned} xc_0 &= 08 * (tp_1 op_0) \\ &\oplus 04 * (tp_2 op_0) \\ &\oplus 02 * (tp_3 op_0) \\ &\oplus tp_1 op_0 \\ &\oplus wc_0 \end{aligned} \quad (7)$$

From the Eq.(7), to implement the inverse MCT 08,04 and 02 Xtime multipliers are used and simplified operation is shown in Table 1.

5.3 QCA implementation of multiplier

According to the literature survey, there are many methods are used in the implementation of XOR gate. In this work, our objective is to design the simplified Xtime multiplier using QCA. Hence, we designed QCA based Xtime multipliers using four different models of XOR gate that are shown in Fig. 7. It was observed that model four design is simple and hence model 4 xtime 02 multiplier is utilized throughout our work.

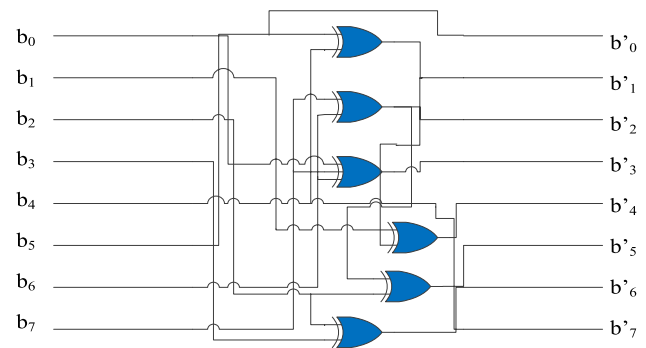


Fig. 6 08 Xtime GF multiplier

Table 1 Multiplier table

Output	08 multiplication	04 multiplication	02 multiplication
b'0	b5	b6	b7
b'1	$b6 \oplus b5$	$b6 \oplus b7$	$b0 \oplus b7$
b'2	$b7 \oplus b6$	$b0 \oplus b7$	b1
b'3	$b0 \oplus b7 \oplus b5$	$b1 \oplus b6$	$b2 \oplus b7$
b'4	$b1 \oplus b6 \oplus b5$	$b2 \oplus b6 \oplus b7$	$b3 \oplus b7$
b'5	$b2 \oplus b7 \oplus b6$	$b3 \oplus b7$	b4
b'6	$b3 \oplus b7$	b4	b5
b'7	b4	b5	b6

6 Results and discussion

6.1 QCA simulation setup

This section presents the comprehensive results of area, power, and delay of the several functional modules that comprise the core component of the AES MixColumn. QCADesigner 2.0.3 simulation tool has been used for layout design and functional verification of the circuit [40]. Another tool QCADesigner-E is used for energy estimation [41]. The simulation engines in QCADesigner v2.0.3 are bi-stable approximation engine and coherence vector engine. In the bi-stable approximation engine, the designer could change the number of samples and the number of iterations per

Fig. 7 Xtime 02 multiplier using different XOR gate

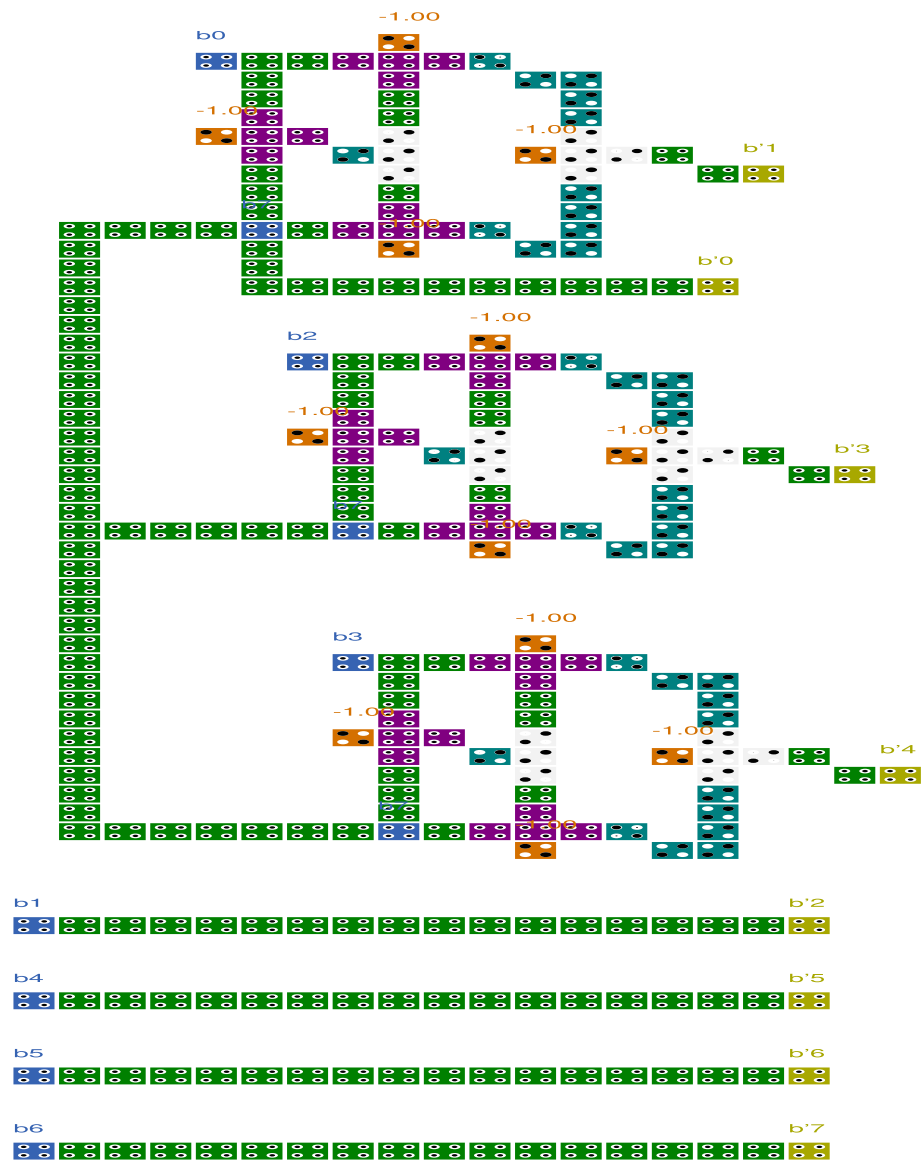
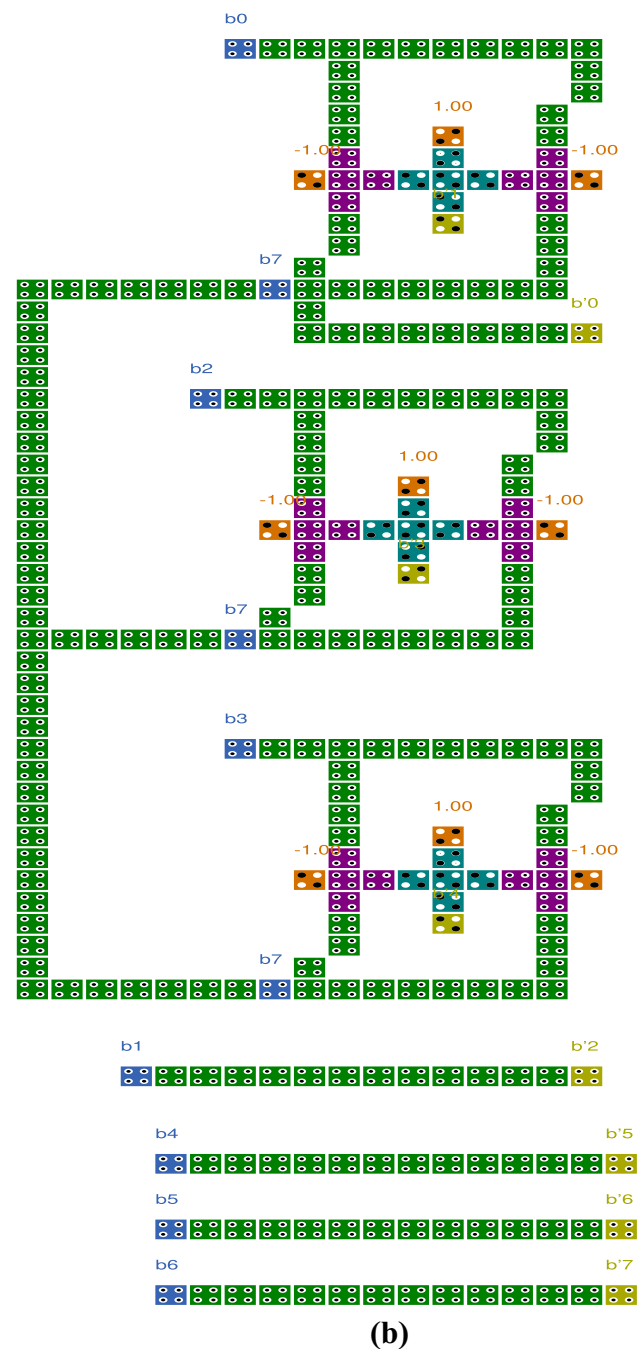


Fig. 7 (continued)



sample, whereas in the coherence vector engine, the designer controls the temperature, relaxation time, total simulation duration, and so on.

The coherence vector model is used in the proposed QCA implementation. In addition, the Euler transformation, randomized simulator, and GAUSS type clock signal were chosen. The values of the simulation setups parameters are shown in Table 2.

Throughout the simulation procedure, we simulate the Type 1-model xtime multiplier in coherence vector energy simulation using a total of 500,000 samples. In the

simulation configuration, a total of eight inputs and eight outputs were utilized. During the steady state simulation, it was observed that the system required seven iterations to converge in the initial steady state experimentation.

6.2 Energy dissipation, quantum cost, area, and latency

The design complexity, cell area, energy dissipation, total area, average energy, quantum cost, and number of clock zones of the proposed QCA-based Xtime multiplier module

Fig. 7 (continued)

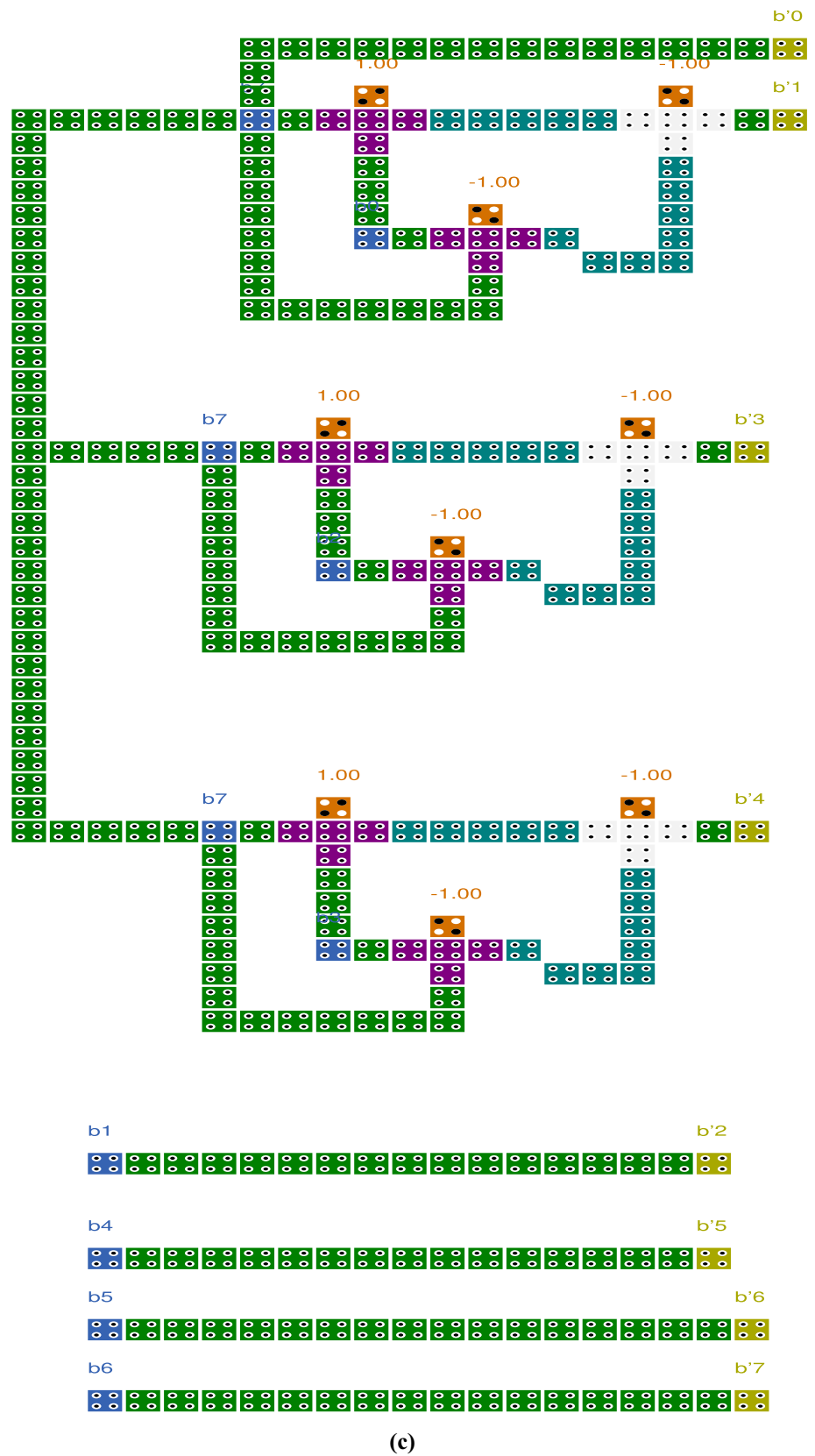
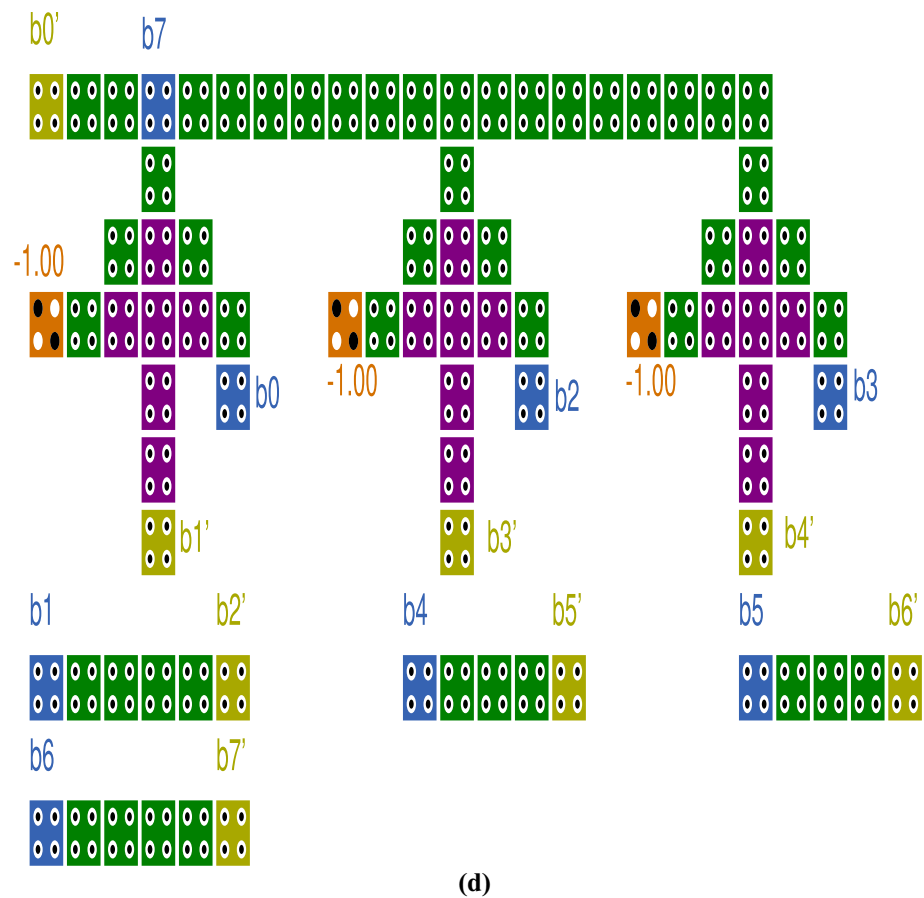


Fig. 7 (continued)



are all evaluated. All of these are divided into four categories: energy, area, clock, and quantum cost.

Table 3 shows the values of these parameters. The amount of cells used in the design defines the design's complexity. The table shows the total complexity of the various designs. The design complexity varies from one design to the next. Total area is another area characteristic determined by the

total rectangular area covered by the QCA cell. The design's cell area is determined as the product of the design complexity and one cell area. A typical QCA cell is 18nm long and 18nm wide. The table lists all four models of Xtime multiplier modules' size and cell area. A cost function is essential in any design that is chosen for implementation. In this case, the cost function is taken into account, which is determined from the system's area and delay value. It is also known as area delay-cost, which is the product of area and delay. The area is in nm^2 and the delay is in clock cycles.

From Table 3, it is observed that Model 4 uses only 31.23% of cells compared to all other models. Additionally, the cell area occupied by Model 4 is only 32.5, 31.2, and 29.38% of Models 1, 2, and 3, respectively. Furthermore, the energy dissipation of Model 4 is 43, 48.8, and 43.3% lower than the other models, respectively. The average energy per cycle of Model 4 is 43.52, 48.9, and 43.52% of Models 1, 2, and 3. Lastly, the quantum cost of Model 4 is reduced by 85.5% compared to the others.

Another critical factor is energy dissipation. It is regarded as the circuit's energy bath. An energy transfer occurs from one cell to another. The energy bath was computed using the QCA designer E tool. The detailed energy dissipation and average energy per cycle of the Xtime multiplier across four

Table 2 Simulation setup parameters

Description/parameters	Value
Temperature	1.000K
Relaxation time	1.000e-015 s
Clock period	4.00e-012 s
Input period	4.00e-012 s
Total simulation time	5.00e-11 s
Clock high	9.800e-22 J
Clock low	3.8000e-023 J
Clock shift	0
Clock slope	1.000e-012 s
Radius effect	80.00nm
Relative permittivity	12.900
Layer separation	11.500mm

Table 3 Xtime multiplier comparison

Parameters	Type 1 model 02 Xtime multiplier	Type 2 model 02 Xtime multiplier	Type 3 model 02 Xtime multiplier	Type 4 model 02 Xtime multiplier
No of inputs	08	08	08	08
No of outputs	08	08	08	08
No of QCA cell	285	269	286	84
No of clock signal	04	04	04	04
Total area (μm^2)	0.52	0.51	0.52	0.075
Cell area (mm^2)	92.340	87.156	92.624	27.216
Energy dissipation	1.16e-001 eV	1.03e-001 eV	1.16e-001 eV	0.503e-001 eV
Average energy per cycle	10.5e-003 eV	9.34e-003 eV	10.5e-003 eV	4.57e-003 eV
No of clocking zone	04	04	04	02
Delay latency (in clock cycles)	1	1	1	0.5
Quantum cost (area-delay product)	0.52	0.51	0.52	0.075

Table 4 Total and average energy dissipation of different Xtime multiplier models

Energy parameters	Model 1	Model 2	Model 3	Model 4
Total Energy dissipation (*e-1 eV)	1.39	0.95	1.23	0.503
Average energy dissipation (*e-2 eV)	1.26	0.87	1.12	0.45

different models were obtained through simulation in the QCA DesignerE tool.

Table 4 list the total and average energy dissipation of the different Xtime multiplier module.

According to the simulation results, Model 1 exhibits a total energy dissipation of 1.39e-001 eV with an average energy dissipation per cycle of 1.26e-002 eV, while Model 2 shows a lower total energy dissipation of 9.59e-002 eV and an average of 8.72e-003 eV per cycle. Model 3 has a total energy dissipation of 1.23e-001 eV and an average per cycle of 1.12e-002 eV. Model 4 shows the lowest energy dissipation, with a total of 5.03e-002 eV and an average energy per cycle of 4.57e-003 eV. Overall, Model 4 is the most energy-efficient, making it a promising candidate for QCA implementation in cryptographic operations.

7 Conclusion and future works

AES is one of the most secure cryptography algorithms in terms of security concerns. Sbox and Mixcolumn are the most dominant power-consuming and complex-function modules. Here, QCA, an energy-efficient alternative to CMOS methods has been proposed for MixColumn operation. A fundamental functional module of Xtime has been implemented in different XOR models to compare the energy efficiency. First, the proposed model has been simplified in the GF(2⁸) function and the same Xtime multiplier has been implemented in different XOR structures for energy efficiency comparison. Indeed, the proposed QCA implementation Model 4 Xtime multiplier

dissipates the minimum amount of energy comparable to other types. The total amount of energy dissipation and average energy dissipation are tabulated and analyzed.

In the future, a complete structure of MixColumn can be implemented, and AES encryption and decryption modules could be implemented in QCA for reversible as well as quantum computing structures.

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Declarations

Conflict of interest All authors have no conflict of interest.

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